

WENRUI HUANG

CONTACT

Email: wenrui_huang@brown.edu
Office: Kassar House 020
Department: Mathematics, Brown University
Address: 151 Thayer Street, Providence, RI 02912

EDUCATION

Brown University <i>Ph.D. in Mathematics</i> • Advisor: Benoît Pausader	Providence, RI 2022 - 2027 (<i>expected</i>)
University of Science and Technology of China (USTC) <i>B.S. in Mathematics</i> • Advisor: Yong Wei • Thesis: Mean curvature flow with surgery and its applications	Hefei, China 2018 - 2022

RESEARCH INTERESTS

Partial differential equations, nonlinear dispersive PDEs and kinetic equations, stability and long-time behavior.

PAPERS AND PREPRINTS

1. *Long-Time Dynamics of Vlasov-Hartree Systems across the Coulomb threshold*, with M. Xie. [arXiv:2608.04402](#).
2. *Scattering map for the Vlasov-Poisson system with a repulsive harmonic potential*, with H. Kwon. [arXiv:2602.21344](#).
3. *Stability and instability of small BGK waves*, with D. Bian, E. Grenier, and B. Pausader. [arXiv:2601.10030](#).
4. *The Vlasov-Poisson System with a Perfectly Conducting Wall: Convex Domains*, with B. Pausader and M. Suzuki. **Commun. Math. Phys.** 407, 219 (2026). [arXiv:2412.13434](#).
5. *Modified wave operators for the Hartree equation with repulsive Coulomb potential*, [arXiv:2411.02737](#).
6. *Scattering of the Vlasov-Riesz system in three dimensions*, with H. Kwon. **Annales Henri Poincaré** (2026). [arXiv:2407.16919](#).

TALKS

- | | |
|--|-----------|
| • GLESPA Seminar, Brown University | Apr. 2026 |
| • Poster Session, Workshop on Kinetic Theory and Fluids, University of Wisconsin–Madison | Mar. 2025 |
| • Brown–Yale PDE Seminar | Apr. 2024 |
| • GLESPA Seminar, Brown University | Apr. 2024 |
| • Brown–Yale PDE Seminar | Nov. 2023 |

TEACHING

Brown University

- TA, MATH 0180 Calculus III, Fall 2026
- TA, MATH 0190 Calculus II (Physics/Engineering), Fall 2025
- TA, MATH 0180 Calculus III, Spring 2024
- TA, MATH 0180 Calculus III, Fall 2023
- TA, MATH 0100 Calculus II, Spring 2023

University of Science and Technology of China

- TA, Riemannian Geometry, Spring 2022
- TA, Differential Geometry, Fall 2021
- TA, Topology, Spring 2021 and Spring 2022